

Detailed Document on Fruit Juice Composition and Effects

Fruit Juice Composition and Effects

1. Ingredients and Percentages

Fruits (60%)- Jackfruits

- **Description**: Jackfruits serve as the primary ingredient in the juice, making up 60% of the total composition.
- **Effects**:
 - **Flavor and Texture**: Jackfruits provide the bulk of the juice, imparting its primary flavor and texture. Their natural sweetness and unique taste are central to the overall flavor profile of the juice.
 - **Nutritional Value**: Jackfruits are rich in vitamins, minerals, and dietary fiber, which contribute to the nutritional value of the juice. They are a good source of vitamin C, potassium, and antioxidants.

Water (20%)

- **Description**: Water is added to the juice to dilute the concentrated fruit content and achieve the desired consistency.
- **Effects**:
 - **Consistency**: Water helps in achieving a smooth and drinkable consistency, making the juice more palatable.
 - **Hydration**: As a vital component, water provides essential hydration to the body, complementing the nutritional benefits of the juice.

Sweetness (10%)- Honey or Maple Syrup

- **Description**: Natural sweeteners like honey or maple syrup are used to enhance the sweetness of the juice without the need for artificial additives.

- **Effects**:

- **Natural Sweetness**: Acts as a natural sweetener, adding a rich, sweet taste to the juice. Honey or maple syrup provides a more wholesome and flavorful sweetness compared to refined sugar.
- **Nutritional Benefits**: Both honey and maple syrup contain antioxidants and beneficial compounds that can contribute to overall health.

Flavorings (5%)- Citrus Fruits

- **Description**: Citrus fruits such as lemon, lime, or orange are used to add a tangy flavor to the juice.

- **Effects**:

- **Flavor Enhancement**: Adds a tangy citrus note, enhancing the overall flavor profile of the juice. This helps balance the sweetness and provides a refreshing taste.
- **Nutritional Boost**: Citrus fruits are rich in vitamin C and other antioxidants, adding additional health benefits to the juice.

To Maintain Color and Taste (3%)- Ascorbic Acid

- **Description**: Ascorbic acid, also known as vitamin C, is used to preserve the color and enhance the taste of the juice.

- **Effects**:

- **Color Preservation**: Ascorbic acid helps maintain the natural color of the juice, preventing it from oxidizing and turning brown.
- **Taste Enhancement**: Enhances the overall taste of the juice by maintaining its freshness and natural flavor.
- **Antioxidant Properties**: Acts as an antioxidant, preserving the freshness of the juice and extending its shelf life.

Preservatives (2%)- Citric Acid

- **Description**: Citric acid is used as a preservative to extend the shelf life of the juice by inhibiting microbial growth.

- **Effects**:

- **pH Reduction**: Lowers the pH of the juice, creating an acidic environment that inhibits the growth of bacteria and other microorganisms.
- **Preservation**: Helps in preserving the juice, ensuring it remains safe for consumption over a longer period.